

Automatic Factor Grouping

Exposure deduplication and latent return factors

Abstract

This cluster-only run groups 140 usable signals without known-factor labels. HDBSCAN first collapses signals that rank the same stocks, including sign-reversed definitions. The resulting primitive FMP returns are then decomposed by parallel-analysis PCA and varimax rotation. The fresh version=20260829T051952Z sample produces 5 final candidate factors and constructs one FMP book from each factor's signed primitive composites. No signal elimination or factor-admission rule is applied.

1 Sample and method

Setting	Value
Gold	version=20260829T051952Z
Requested symbols	all
Panel symbols	3014
Exposure window	2012-01-01 to 2022-12-31
Raw price floor	\$1.00
Market-cap screen	top 90.0% of names
Financials excluded	true
Minimum names / usable month	20
Minimum usable months	24
HDBSCAN minimum cluster size	2
Block-bootstrap repetitions	10
Primitive FMP sector demeaning	sector
Primitive FMP beta neutrality	false
Final factor FMP sector demeaning	sector
Final factor FMP beta neutrality	true
Parallel-analysis repetitions	10
Minimum absolute rotated loading	0.30
Minimum primitives / final factor	2

Monthly cross-sectional Spearman correlations are overlap-weighted through time. Exposure distance is $1 - |\rho|$, so a signal and its negation are treated as the same measurement and their sign is stored separately. HDBSCAN is used only for redundancy reduction. Final factors come from the common 131-month primitive-return window (2012-02-01 to 2022-12-01); no external factor return enters either stage. Beta is deliberately left in primitive returns so beta/BAB can appear as a latent risk dimension; beta neutrality belongs to later portfolio construction, not unsupervised grouping.

2 Exposure-redundancy primitives

HDBSCAN produced 85 primitives: 41 multi-signal groups and 44 singletons. This is not treated as the final factor count; the earlier HDBSCAN-only interpretation failed because a dense redundancy map naturally preserves many unrelated signals as noise.

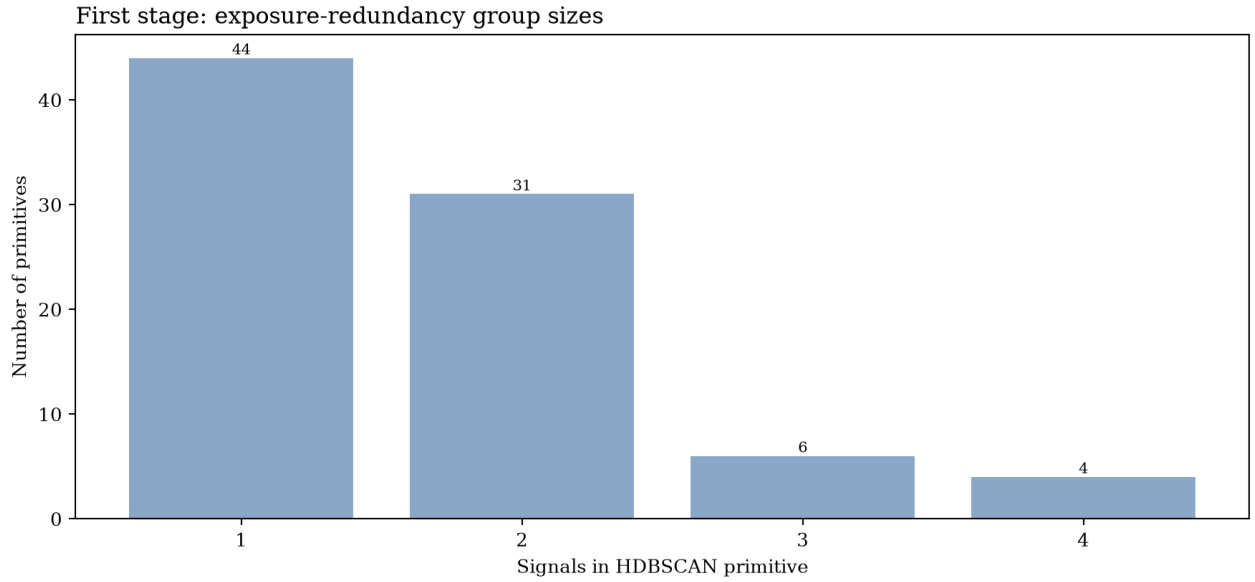


Figure 1: Distribution of signal counts in the exposure-redundancy primitives.

3 Latent factor count

Parallel analysis compares observed correlation-matrix eigenvalues with the 95th percentile from independently circular-shifted primitive returns. It retains 5 dimensions; 5 remain after the loading and minimum-signal rules.

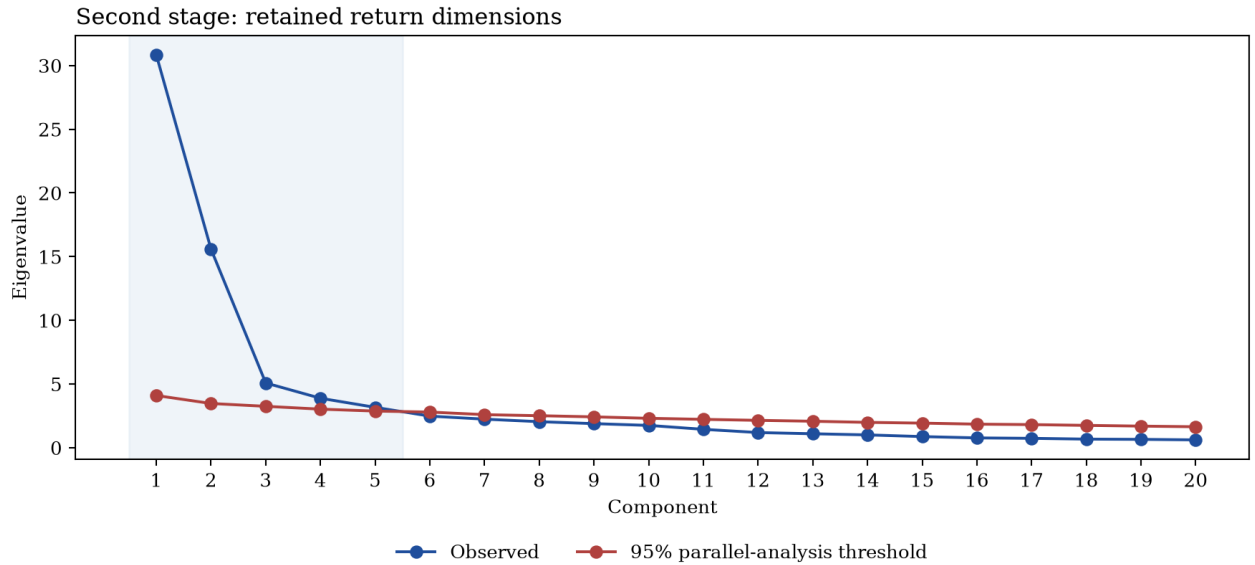


Figure 2: Observed and simulated eigenvalues. The shaded components are retained.

4 Rotated structure

Varimax rotation makes each primitive load mainly on one component. Loading sign orients a primitive within its factor; loading magnitude is not used as a portfolio weight. Every final factor gives each oriented primitive composite one equal vote, regardless of how many raw signals that primitive contains.

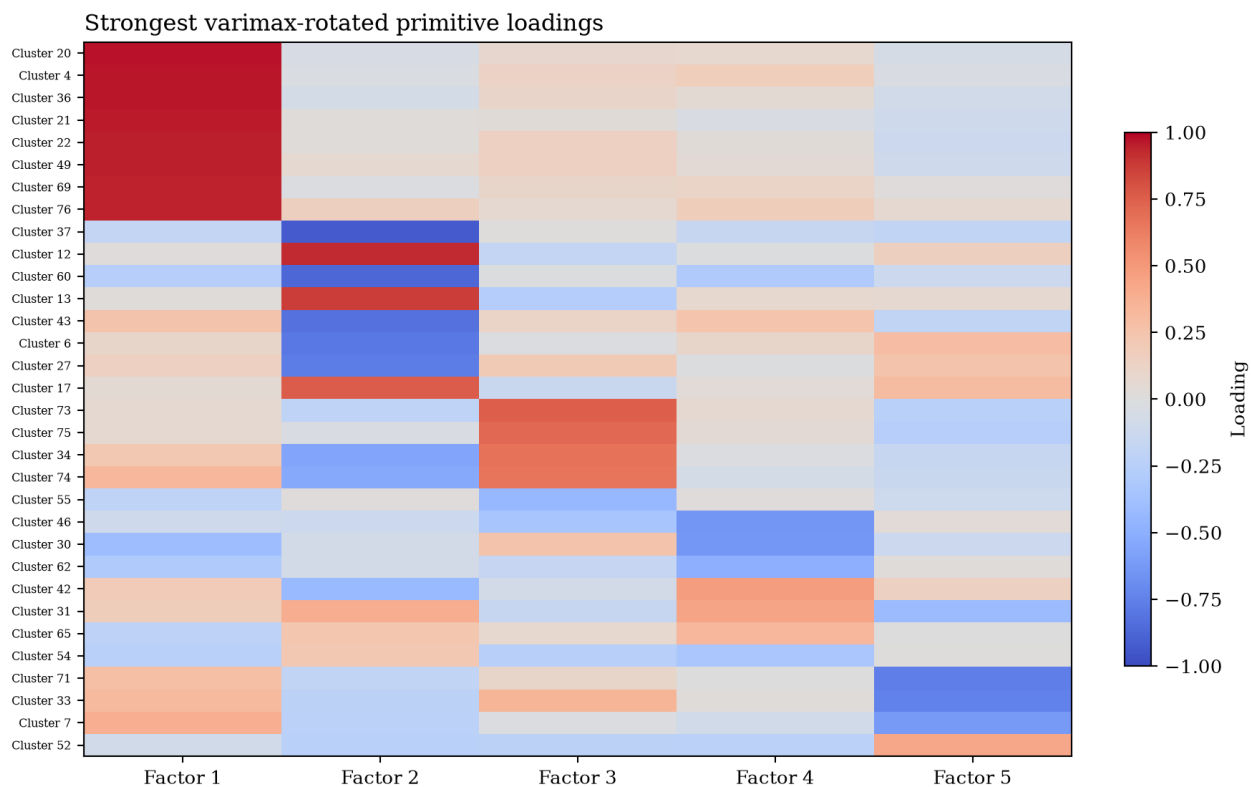


Figure 3: Rotated loading matrix, showing up to eight strongest retained primitives per factor.

5 Final candidate factors

The signs below give orientation only. Economic quality has not been used to remove any signal or factor in this report.

Factor	Primitives	Raw signals	Signed primitive groups	Final signed signals
Factor 1	42	77	+Cluster 20, +Cluster 4, +Cluster 36, +Cluster 21, +Cluster 22, +Cluster 49, +Cluster 69, +Cluster 76, -Cluster 10, +Cluster 18, +Cluster 77, -Cluster 79, +Cluster 59, +Cluster 32, +Cluster 23, -Cluster 45, -Cluster 58, -Cluster 82, -Cluster 81, +Cluster 50, +Cluster 16, +Cluster 63, +Cluster 64, -Cluster 66, +Cluster 29, +Cluster 48, -Cluster 11, +Cluster 25, +Cluster 19, +Cluster 5, -Cluster 1, -Cluster 3, +Cluster 61, -Cluster 2, -Cluster 15, +Cluster 14, +Cluster 9, -Cluster 40, +Cluster 44, -Cluster 38, +Cluster 41, +Cluster 83	+bab 252v 1260c FRENCH CRSP, +bab shrunk 252v 1260c FRENCH CRSP, -beta 252v 1260c FRENCH CRSP, -beta shrunk 252v 1260c FRENCH CRSP, +bab 252v 1260c SPY, +bab shrunk 252v 1260c SPY, -beta 252v 1260c SPY, -beta shrunk 252v 1260c SPY, +bab 252v 1260c VTI, +bab shrunk 252v 1260c VTI, -beta 252v 1260c VTI, -beta shrunk 252v 1260c VTI, +operating profitability, +roa, +roe, +rota ebit, +accruals, +accruals percent, +accruals total, +eps growth yoy, +net income growth yoy sign flip, -ivol 21d, -lottery max 5 21d, -vol 63d, -amihud illiq 21d, +dollar volume 21d, +buffett ronta proxy, +roc, -cash ratio, -cash to assets, +corr 1260d FRENCH CRSP, +corr 1260d VTI, +earnings yield, +ebit to ev, +ebit growth yoy ttm, +net income growth yoy ttm, +ebit margin ttm, +net income margin ttm, +fcf, +fcf margin, +fcf margin ttm, +fcf to equity, +fcf to market cap, +fcf yield, +gp to equity, +gpoa, +growth roa 5y, +growth roe 5y, +log market cap, +market cap, +net margin, +operating margin, -rd to opex, -rd to revenue, -ret volume corr 63d, -skew 63d, +surprise, +surprisePercentage, +asset turnover, -beta stability 252d 252v 1260c SPY, +cash earnings ttm, +cfoa, +corr 1260d SPY, -equity issuance yoy, +ev, +fcf growth yoy ttm, +growth cfoa 5y, +high 52w 252d, -interest to debt, +is positive net income, +net income to market cap, +net operating assets proxy, -roe volatility, -sga to revenue, -share turnover 21d, +is negative equity
Factor 2	23	37	-Cluster 37, +Cluster 12, -Cluster 60, +Cluster 13, -Cluster 43, -Cluster 6, -Cluster 27, +Cluster 17, +Cluster 24, -Cluster 28, -Cluster 35, +Cluster 8, +Cluster 68, +Cluster 85, -Cluster 26, +Cluster 67, -Cluster 72, -Cluster 80, -Cluster 51, +Cluster 47, +Cluster 57, +Cluster 84, -Cluster 39	-accruals noa proxy, -asset growth yoy, -equity growth yoy, +debt to assets, +debt to equity, -ncav to mc, +assets to mc, -ev to assets, +book to market, -pb, +earnings to price, -pe ttm, +gp ev, +gross profit to ev, -gross margin, -gross profit margin ttm, -gross profit growth yoy ttm, -revenue growth yoy, -growth gmar 5y, -growth gpoa 5y, -momentum 21 756, +value proxy 1260d, -ps ttm, +sales to price, -resid ret 252v 1260c SPY, -ret zscore 63d, -altman z, +capital intensity, -current ratio, +equity growth yoy sign flip, -evs ttm, +interest to opinc, +is negative net operating assets, -momentum 126 252, -roic nopat, +is negative ev, +is negative invested capital

Factor	Primitives	Raw signals	Signed primitive groups	Final signed signals
Factor 3	5	6	+Cluster 73, +Cluster 75, +Cluster 34, +Cluster 74, -Cluster 55	+momentum 21 252, +resid momentum 21 252, -dso, +momentum 21 126, +momentum 21 252 vol 63d, +momentum 21 63
Factor 4	7	9	-Cluster 46, -Cluster 30, -Cluster 62, +Cluster 42, +Cluster 31, +Cluster 65, -Cluster 54	-is negative working capital, +working capital proxy, +is positive income tax, +tax rate, +accruals wc proxy, -capex to revenue, -dpo, -growth acc 5y, +info discreteness 21 252
Factor 5	4	7	-Cluster 71, -Cluster 33, -Cluster 7, +Cluster 52	-bb pctb 20d, -rsi 14d, -stoch k 14d, -ma dist 63d, -price zscore 63d, +debt issuance yoy, -momentum 0 21

6 Final factor FMP performance

Each factor is constructed once from its signed primitive composite exposures using the configured rank transformation, composite-level sector demeaning, signal-weighted long-short portfolio and factor-level beta neutrality. The curves and statistics use the same complete common return window.

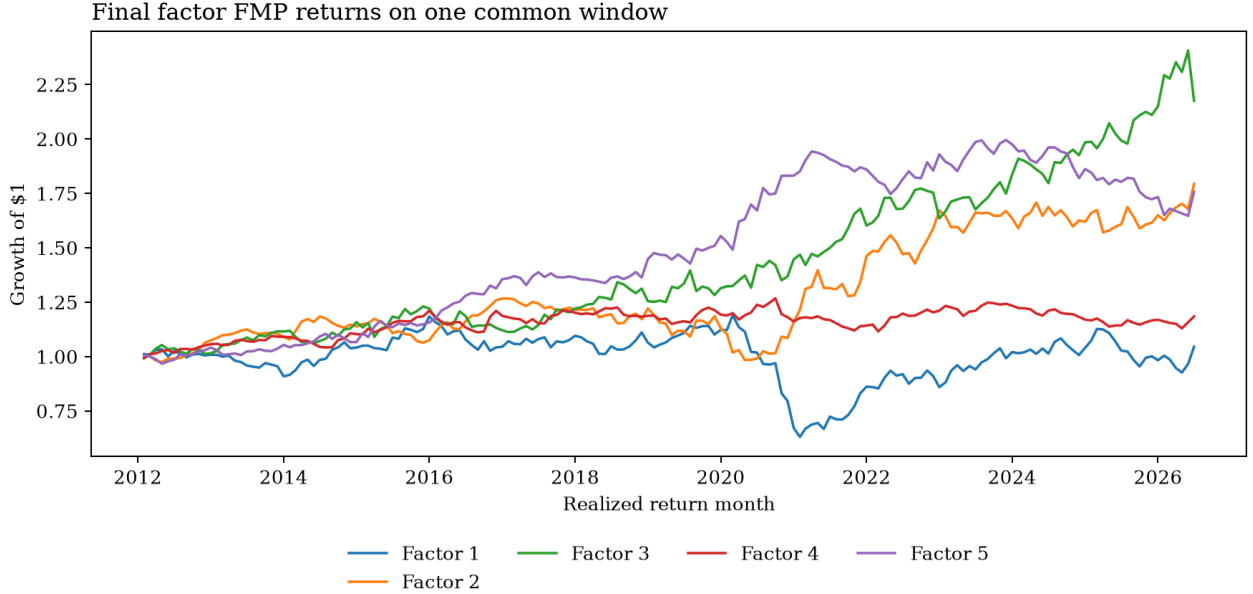


Figure 4: Net cumulative returns of the final factor books.

Factor	N	Start	End	Ann. return	Ann. vol.	Sharpe	HAC t	Max DD
Factor 1	174	2012-02-01	2026-07-01	0.98%	11.58%	0.09	0.29	-47.02%
Factor 2	174	2012-02-01	2026-07-01	4.44%	9.05%	0.49	1.69	-22.38%
Factor 3	174	2012-02-01	2026-07-01	5.77%	8.94%	0.64	3.31	-9.63%
Factor 4	174	2012-02-01	2026-07-01	1.29%	4.92%	0.26	1.06	-11.99%
Factor 5	174	2012-02-01	2026-07-01	4.12%	6.81%	0.60	2.42	-17.47%

7 Interpretation and reproducibility

This remains an exploratory return-risk taxonomy: the performance table describes the resulting books but does not select or eliminate factors. Exposure discovery run 20260904T151449Z_c44b2a4e1c. The run directory preserves primitive definitions, raw-signal lineage, rotated loadings, stock-level final exposures, primitive and final FMP returns, configuration and manifest for Factor Runner or HMM reuse.